

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template AT - 15 points

Guidance before filling out the template on the next page:

Because you will iterate (repeat) your design multiple times, you will need to fill out multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. Iteration must shine through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision-making on which design to include with a design matrix. Create a plan of work, ideally using a Gantt chart that may be housed in this document or in a different day-to-day document. Generate a design blueprint with enough details that someone else not familiar with your project could also build your design.

You will then proceed forward to build and test this design, returning to Element D afterward when you are ready to create more design ideas. Fill out each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Name: _____ Date: _____ Class: _____

<i>Design Idea #1</i>	<i>Class Organizer</i>
Date:	2/26/26
Description:	<i>A classroom organizer that has 5 slots, intended to hold a tissue box, hand sanitizer, bandages, pencils, and bathroom passes.</i>
Drawing that includes distinguishing information such as scale, dimensions, and materials:	<i>14 inches in length</i> <i>12 inches wide</i> <i>6 inches height + 4 inches for handle height</i> <i>3D printed using PLA filament</i>

<i>Design Idea #2</i>	<i>Marker Holder</i>
Date:	2/26/26
Description:	<i>A marker holder designed to sit on the whiteboard ledge.</i>
Drawing that includes distinguishing information such as scale, dimensions, and materials:	<i>7 inches in length</i> <i>7 inches depth/width</i> <i>5 inches in height</i> <i>3 inches height for the front section (eraser holder)</i> <i>1 inch for hook depth</i> <i>2 inches for hook height</i> <i>3D printed using PLA filament</i>



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<i>Design Idea #3</i>	<i>Drawer</i>
Date:	<i>2/26/26</i>
Description:	<i>Holding any types of materials that are storageable.</i>
Drawing that includes distinguishing information such as scale, dimensions, and materials:	<i>14 inches in hight 7 inches width 8 inches in length 3D printed with PLA filament</i>

<i>Design Idea #4</i>	<i>General organizer</i>
Date:	<i>2/26/26</i>
Description:	<i>Has 4 compartments, tissues, hand sanitizer, pencils, hall passes</i>
Drawing that includes distinguishing information such as scale, dimensions, and materials:	<i>12x6 inches, height is 6 inches and should be 3d printed with PLA.</i>



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Design Idea Comparison and Selection (*D1, D2*)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options.

A decision matrix is used to allow for systematic, weighted comparisons.

If using weights in a decision matrix, be sure to explain and justify the weighting based on your design requirements.

You may use the weighted design matrix template below, inserting your criteria and design options.

(List your requirements for Element C below.)	Weight (multiplied by each design score)	Design 1	Design 2	Design 3	Design 4
Compact	5	2	4	1	3
Versatile	3	5	3	5	4
Simple design	4	3	5	1	3
Easy to use	4	4	5	3	4
Durable	2	3	4	2	3
Total score	N/A	50	900	180	960

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Our team chooses design _____ to solve the _____ problem
because _____
(which specific requirements does it fill).

Design Blueprint (D4)

Describe all of your design details so that someone unfamiliar with your project can build your prototype.

Include orthographic sketches (front, top, and side views), dimensions (include units), CAD drawings (if taught in your class).

What manufacturing methods, material choices, supply lists, and costs do you anticipate for the prototype?

When you iterate on this idea, you will copy the design blueprint section for the second design.

Design Blueprint #1

Date:

Sketch or CAD Drawing (orthographic/multiview):

Manufacturing Methods (cut, saw, paste, screw, tie, glue):

Step 1:

Step 2:

Step 3:

Supply List and Costs:

PLA Filament

Name: _____

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Item	Cost each	# needed	Total Cost
Class Organizer (Design 1)	\$20	2	\$40
Marker Holder (Design 2)	\$15	4	\$60
Drawer (Design 3)	\$30	2	\$60
General Organizer (Design 4)	\$20	2	\$40
Totals	\$85	10	\$850

Plan of Action (D3)

- A Gantt chart is encouraged to outline the primary steps expected on a visual timeline. [Here is one you can use.](#)
- Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document.
- The Gantt chart will be a future assignment so if you start now you will be ahead.
- Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.
- Make screenshots of the document instead of using the link.



Wait until you create and test your first prototype to fill out the next section.

Iterations for the Solution (How did you change your prototype after building and testing it?):

Date of Iteration:

Description of Iteration:



Name: _____

Date: _____

Class: _____

Images if applicable:

Date of Iteration:

Description of Iteration:

Images if applicable:

Date of Iteration:

Description of Iteration:

Images if applicable:

Rubric

Element D: Design Concept Generation, Analysis and Selection						
	5	4	3	2	1	0
1. Process for Generating & Comparing Possible Solutions. The Process for	was <i>comprehensive,</i> <i>iterative,</i> <i>and</i> <i>consistently</i>	was <i>thorough,</i> <i>iterative,</i> <i>and</i> generally defensible	was <i>adequate,</i> <i>generally</i> iterative and defensible <i>(viable design</i>	was <i>partial or</i> <i>overly general</i> <i>and only</i> <i>somewhat</i> iterative and/or	was <i>incomplete</i> <i>or minimally</i> <i>iterative</i> <i>and/or</i> <i>defensible</i>	<i>was not evident</i> <i>that there was</i> <i>an attempt</i>

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Generating Possible Solutions _____.	defensible (viable design highly likely)	(viable design likely)	<i>is possible</i>	defensible (raising issues with the design viability)	<i>(raising issues with the design viability)</i>	
2. Design Solution Choice. The Design Solution chosen _____.	was <i>well-justified</i> and demonstrated attention to <i>all</i> design requirements	<i>was justified</i> and demonstrated attention to <i>most if not all</i> , design requirements	<i>was explained with reference to at least some</i> design requirements	<i>was not sufficiently explained with reference to</i> design requirements	<i>lacked support related to</i> design requirements	<i>was not attempted or evidence not provided</i>
3. Plan of Action. The plan of action _____.	<i>has considerable merit and would easily support repetition and testing for effectiveness by others</i>	<i>would support repetition and testing for effectiveness by others</i>	<i>might not clearly or fully support repetition and testing for effectiveness by others</i>	<i>has insufficient detail to allow for testing for replication of results</i>	<i>is lacking to the point that the design solution cannot be tested</i>	<i>cannot be executed or does not exist</i>

